Digital Twin Vision for Freight Railroads: A Case Study in North America

Silvia Galván-Núñez, Ryan Alishio, and Christopher Johnson

***Abstract*— In recent years, the concept of "digital twins" (DTs) has gained increasing attention in the railroad industry. This complex concept allows the creation of digital representations of physical assets or processes and can support a railroad's transportation system throughout its lifecycle. For decades, MxV Rail has used the Facility for Accelerated Service Testing (FAST (Registered trademark.) to enable research and testing designed to improve the safety, reliability, and efficiency of North American railroads. At FAST, data streams generated by 1) operations, 2) maintenance, 3) wayside and onboard detector technologies, and 4) individual testing are employed to better understand the effects of heavy axle loads (HAL) as high tonnage accumulates on track components at various stages of wear. The development and implementation of DTs have strong potential to support decision-making for infrastructure and mechanical systems with regard to monitoring, predictive maintenance, and knowledge transfer to conditions where data is unavailable. This paper seeks to open the conversation by reviewing the concept of DTs in the North American railroad industry and showing the vision of the implementation layers for making a DT for FAST and the current efforts to achieve this vision.**

***Index Terms*— Digital engineering, digital twins, railroad freight, FAST.**

I. Introduction

With the growing interest and rapid advancements in digital transformation, along with the drive to create more value from both large and small data streams, the concept of digital twins (DTs) has emerged as a focal point across various industries.

Before diving into the concept of DTs, it is necessary to introduce the concept of digital engineering (DE), which is defined as a comprehensive approach that consolidates information throughout the lifecycle of a megaproject. This approach leverages digital information management systems for designing, engineering, constructing, and maintaining complex projects [1]. When implemented properly in the aerospace, automotive, energy, and other industries, DE can significantly improve the quality of services and achieve notable cost reductions in these industries [1].

In addition to DE, DTs represent a concept with diverse interpretations. With origins at the National Aeronautics and Space Administration (NASA), a DT is defined by [2] as "*a set of virtual information constructs that fully describes a potential or actual physical manufactured product from the micro atomic level to the macro geometrical level. At its optimum, any information that could be obtained from inspecting a physical manufactured product can be obtained from its digital twin.*" Both DE and DT can be considered the synthesis of established methods and processes, forming a unified framework to improve system management across various sectors. Whether the systems exist or are hypothetical, this framework may integrate techniques such as physics-based modeling, machine learning, and analytics to create cohesive digital system replicas. While some researchers emphasize the real-time visualization capabilities of DTs to distinguish them from traditional modeling and simulation methods, other researchers point to the creation of a virtual duplicate of a physical object as the key innovation for a DT [3]. The concept of DTs has been implemented across a variety of fields and industries, including manufacturing [3], [4], automotive [5], aerospace [6], [7], and energy [8], [9], among others.

Based on various applications and definitions found in literature, the conceptualization of DTs has expanded from representing only industrial objects or products to representing almost any physical or intangible object as well. Most published work suggests that the definition of a DT may vary depending on the application area. For instance, in the initial product design phase, a physical object may not be required for the creation of a DT, but it may be essential for asset management. This perspective reinforces the need for a clearly defined DT and invites the establishment of common ground for different industries and application areas. As presented later in this paper, the railroad industry has made multiple efforts to define and apply the concept of DTs in the context of structural health monitoring of track components, asset management, and mechanical systems. However, as in other sectors, there is still no unified concept, indicating the possibility of ongoing opportunities for further exploration and definition within the industry. The Facility for Accelerated Service Testing (FAST®) provides continuous monitoring, research, and testing of track and mechanical systems for the North American railroads and offers excellent opportunities to explore the DT concept.

The objectives of this paper are to 1) continue the conversation regarding exploration of DTs in the North American railroad industry with the goal of defining DTs and 2) present an initial attempt to create a DT for the FAST loop. The remainder of the paper is organized as follows: Section II

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describes some of the existing literature regarding the use of DTs in railroad; Section III presents an overview of FAST; Section IV offers a vision for the implementation layers needed to make a DT for FAST; and Section V covers the concluding remarks.

II. Literature Review

Recently, there has been a growing interest in applying DTs within the railroad industry. This section provides a summary of the literature on DTs in railroads. The review compiles peer-reviewed articles, conference papers, and master's theses that include terms related to DTs, DE, and railroads in their titles, abstracts, and keywords.

On a high level, [10] discusses how DTs can improve the railroad industry by optimizing maintenance planning, design, and operations. In addition, to improve risk analyses, real-time asset health visualizations designed to decrease the time needed to complete technical checks display accurate train makeup to improve efficiency in transshipment preparation, etc. Reference [11] presents a definition for DTs along with an execution plan and the applications of the plan based on stakeholders' questions regarding railcar information. Reference [12] presents an overview of the current state of DT in railroads, highlighting the fact that the emphasis has been placed in the infrastructure side with limited applications in the rolling stock side. Reference [12] also mentions that the status of DT implementation is in the early stages, suggesting a future focus on predictive analytics, lifecycle management, and real-time performance monitoring. Reference [13] provides a detailed literature review and the state of DTs and the railroad industry.

On the mechanical side, [14] considers a real-time freight car braking system designed to identify faults. This study uses both physical and model-based information, employing complex equations to replicate specific parts and integrating them with specialized sensors to enhance accuracy. On the infrastructure side, [15] discusses opportunities for and challenges in automatically generating geometric DTs for track structures. In terms of structural health monitoring, [16] provides applications in bridge health, employing machine learning, the Internet of Things, and cloud technology to enhance monitoring, analysis, and decision-making for bridge maintenance. Another bridge-related study uses machine learning and three-dimensional (3D) simulation to conduct continuous online damage detection on bridges using train-induced responses [17]. For rail yard monitoring, a DT was developed to establish 1) optimal train paths, 2) train make-up, and 3) task orders, with the authors using feedback and corrections from controllers to fine-tune their models [18]. A 3D DT to monitor the structure of railroad tunnels was also created [19]. Additionally, DT applications in track maintenance are explored in a case study on the Passenger Rail Agency of South Africa by [20], demonstrating the practicality and potential of DTs in the railroad industry. Reference [21] depicts the application of a DT to predict rail surface defects, such as rolling contact fatigue in heavy haul operations, by integrating longitudinal train simulation, multibody simulation, and traction modeling.

For grade crossing safety, there is an application for using a DT focused on improving grade crossing safety, concentrating on reducing waiting times for pedestrians and increasing train speeds [22]. Additionally, applications in special trackwork include the creation of a DT to monitor the temperature in turnouts to improve defect detection and maintenance work under different weather conditions [23]. Reference [24] covers the development of a DT to monitor switch machine operations.

Reference [25] presents a DT prototype of a consist to aid in the understanding of the derailment risks for autonomous train operations (ATO) [25], and [26] presents the application of a DT for the European Train Control System for traffic management. For sustainability applications, [27] used the DT concept to assess the sustainability metrics for train station buildings, with a focus on energy efficiency and environmental impact.

Based on the literature reviewed, it is apparent that the usage of the term "digital twin" focuses on rail infrastructure, particularly the structural health monitoring of special trackwork and bridges, with fewer mentions in the mechanical and operational sectors. Most studies using DTs are centered around case studies or remain in the hypothetical and prototype stages, leaving the scalability of DTs not fully explored. The reviewed literature also highlighted implementation challenges, such as data accuracy and automation. Despite these challenges, the increasing interest within the railroad industry is promising, opening up further opportunities for exploration and application.

III. Case Study: The Facility for Accelerated Service Testing

A. FAST Overview

MxV Rail, a wholly owned subsidiary of the Association of American Railroads (AAR), operates FAST as part of the AAR Strategic Research Initiatives (SRI) program. Since operations began in 1976, FAST has evolved into an industry-leading facility for testing railroad infrastructure and mechanical components, maintenance practices, and inspection technologies [28], [29]. FAST focuses on understanding the effects of high tonnage and heavy axle loads on accelerated track degradation and mitigation strategies, as well as investigating the root causes of events to ultimately help improve track and mechanical component performance to reduce risk and increase safety.

At the FAST loop, testing is conducted with an 18,000-ton train of cars loaded to 315,000 pounds, operating in a highly controlled environment while accumulating tonnage of approximately 140 million gross tons (MGT) annually, exceeding that of all but the busiest Class 1 main lines (Fig.1). In 2023, the new FAST loop was built at PuebloPlex in Pueblo, Colorado, United States, and it has been fully operational since November 2023. Spanning nearly 16,000 acres, PuebloPlex is a planned redevelopment of the soon-to-be-former Army Depot. This project will incorporate industries including manufacturing, warehousing, and distribution operations, and rail-related business, among others [30]. The new FAST loop

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Fig. 1. Aerial view of FAST.

consists of 2.8 miles of mainline track designed to enhance research and testing capabilities for track and mechanical components (Fig. 1). FAST includes a wide variety of test components, including:

- ties and fastening systems;
- ballast;
- rail;
- rail neutral temperature monitoring;
- one-way low speed (OWLS) crossings;
- insulated joints;
- electromagnetic acoustic transducer (EMAT) in-motion wheel inspection systems;
- alternative broken rail detection systems;
- track settlement;
- instrumented wheelsets (IWS);
- high-performance wheels;
- wheel profile testing;
- Several others.

B. FAST Data Management

The in-house FAST Internal Records for Statistics and Testing (FIRST) system was designed and developed as part of the initiative to enhance the data collection, storage, and reporting of infrastructure and mechanical system inspections, maintenance, and tests at FAST. This system was modeled using a framework similar to what is typically used for the North American railroads' data management systems and tailored to meet the specific needs of FAST, enabling robust research and testing for North American railroads. Data is gathered from various sources, including wayside detectors, such as hot bearing detectors (HBD), hot wheel detectors (HWD), dragging equipment detectors (DED), truck performance detectors (TPD), automatic equipment identification (AEI) detectors; train operations; tests and measurements scheduling; and rail temperature, etc.

The data collection cycle within the FIRST system begins with test planning before progressing through train operations and the inspection and maintenance of infrastructure and mechanical components. Throughout this process, the tonnage and mileage data are accumulated for both the track and mechanical components. After the process is complete, the system generates reports tailored for various organizational levels.

IV. Digital Twins Vision for Fast

As mentioned in Section II, FAST provides substantial research and testing support for North American railroads.

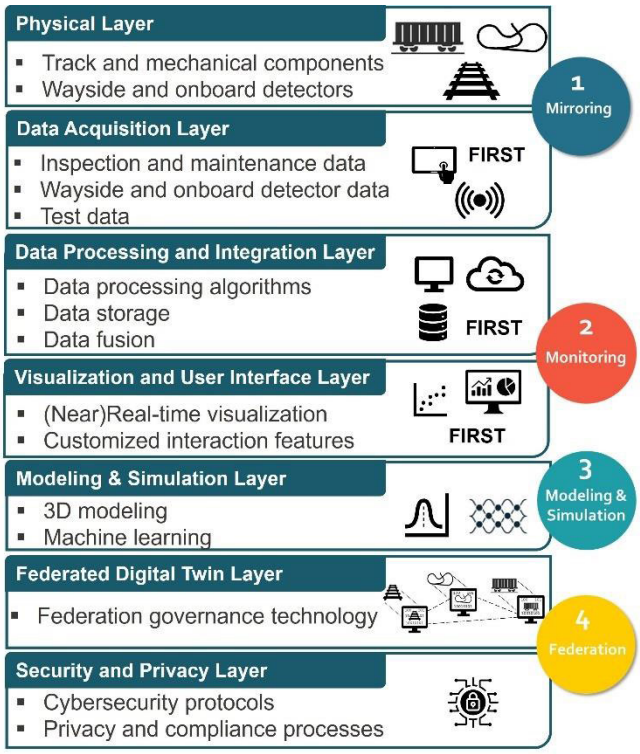


Fig. 2. Digital twin implementation levels and layers for FAST.

Given the diversity of the available data stream sources and the continuous monitoring of the data stream, it is logical to establish a vision for developing DTs for FAST. This initiative is motivated by the attainment of various benefits, such as long-term improvements in train operations, safety, and sustainability. Additionally, DTs can enhance research capabilities through data integration and modeling and aid in the lifecycle management of infrastructure and mechanical components. Ultimately, this effort can also enhance data-driven decision-making.

There are different ways to present the DT implementation pathway. In this paper, the authors present simple, yet complex, implementation levels for DTs. Level 1, mirroring, duplicates a physical object or process as a DT. Level 2, monitoring, is the process of observing and controlling the physical object based on the DT. Level 3, modeling and simulation, involves improving the physical object using 3D modeling, machine-learning models, and other models. Level 4, federation, refers to connecting or merging multiple DTs. For example, in complex systems, such as railroads, multiple DTs are needed to represent part of the system. Level 5, autonomous, refers to the full automation of the DT. Achieving all levels depends on the specific application, e.g., a DT can make it to the modeling and simulation level (level 3) and may not need to get to the federation and/or autonomous levels (levels 4 and 5). To achieve the DT vision for FAST, the authors propose the following implementation levels and layers (Fig. 2).

A. Level 1: Mirroring

There is an interest in building DTs to enhance FAST inspection, maintenance, and research. The physical layer,

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essential for designing and implementing DTs for FAST, includes track/mechanical components and wayside and onboard detectors. Some examples of physical objects that could benefit from DTs include special trackwork components, such as insulated joints and OWLS crossings and switches, as well as bridges, ties, fasteners, and rails, etc. On the mechanical side, the physical objects might be comprised of wheelsets, bearings, and more. These components will be monitored and inspected using data collection systems that may include wayside and onboard detectors, visual inspections, maintenance activities, and testing.

B. Level 2: Monitoring

The monitoring level refers to processing, fusing, storing, and visualizing the data streams associated with the physical objects mirrored by DTs. This step is crucial and challenging, as data sets gathered from multiple sources must be integrated and stored in a way that allows them to be used collectively for analysis. Additionally, the data must enable the generation of visualizations and reports that are actionable at various organizational levels, such as the operations, research and development, and engineering teams that support FAST.

C. Level 3: Modeling & Simulation

The modeling and simulation level involves developing models to 1) enhance the understanding of physical object behavior, 2) support research capabilities, and 3) ensure the safety of train operations. Depending on the specific questions that need to be addressed regarding the physical object, these models may incorporate machine-learning algorithms, finite element analysis, multi-body dynamic simulations, and traditional statistics.

D. Level 4: Federation

Railroads are complex systems, and creating a DT to represent entire railroad networks is not feasible. This complexity also applies to FAST. Therefore, to achieve comprehensive coverage of all operations around the loop, it is necessary to create and interconnect multiple DTs while enforcing cybersecurity protocols to ensure the privacy and security of the collected data. Although the Security and Privacy layer is introduced in Level 4, it is crucial to emphasize that cybersecurity must be considered at every stage of the DTs' design, development, and deployment.

V. Preliminary Results

Regarding the current implementation stage of a DT for FAST, the focus has been on improving the quality and accuracy of the models to ensure they reflect the physical reality of the track and mechanical components tested. This effort has involved refining the algorithms that process operational/maintenance data and improving the system's real-time monitoring capabilities. These improvements have been instrumental in advancing the development of the DT, particularly in terms of mirroring and monitoring, Level 1 and Level 2, respectively, of the DT implementation.

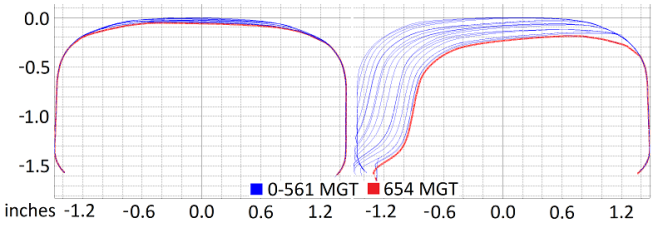


Fig. 3. Example of typical wear patterns for the low and high rail in a 5-degree curve at FAST [30].

Task ID	Task Date	Work Type	Cause	Rail Side	Start Section	Start Subsection	Start Tie
2008	1/30/2025	Weld Removal	Failure	Inside Rail	2	1	18
2007	1/30/2025	Rail Removal	Failure	Inside Rail	1	1	10
2006	1/29/2025	Tie & Fastener Changeout		N/A	15	1	27
1291	1/8/2025	Weld Removal	Maintenance - No Defect	Outside Rail	5	11	9

Fig. 4. Example of FAST track maintenance data recording.

Continuous monitoring for track and mechanical systems is operational alongside the data collection systems, such as the FIRST system. As described in Section IV, the physical layer exists from the FAST loop, the locomotives and cars, wayside detection systems, etc. Track side inspection data includes non-destructive evaluation (NDE) for rail and welds, visual track inspection, rail break detector, and rail profile measurement data, among others. As an example of the track side inspection systems, the NDE for rail and welds allows the identification of internal defects, and, depending on the condition, the defect can be removed from the track immediately or monitored over time for growth and possible removal in the future. Depending on the type of defect, the growth rate is calculated and plotted against the tonnage of that defect in track, measured in MGT. Fig. 3 provides an example of using rail wear data for inspections at FAST [31]—the image on the left presents the typical wear pattern on the low rail in a 5-degree unlubricated curve, and the image on the right shows the wear pattern on the high rail. The blue lines on the rail profile illustrate multiple wear conditions between 0 and 654 MGT.

The data to support the rail wear analysis above come from multiple sources, including, but not limited to, the recording of track maintenance work done through the FIRST system. Fig. 4 presents an example for some of the maintenance summary screens at a tie level. Train operations data is also as important for this analysis to calculate the MGT accumulation, and this data is also tracked in the FIRST system.

On the mechanical side, inspection data includes TPD, HBD, HWD, DED, NDE for wheelsets, and visual inspection of cars and locomotives, among others. For example, TPD data provides near real-time information about the vertical (V) and lateral wheel/rail contact forces (L), L/V ratio, alongside HBD data that can provide insights regarding the condition of the wheelset and bearings so maintenance can be prioritized. Fig. 5 shows a visual representation of TPD data for a day of train operations with the minimum, maximum, and mean for the TPD location on tangent track at FAST. With regard to

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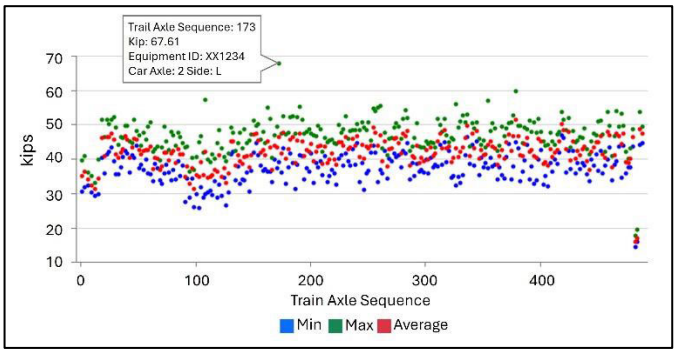


Fig. 5. Example of FAST TPD data at the wheel level.

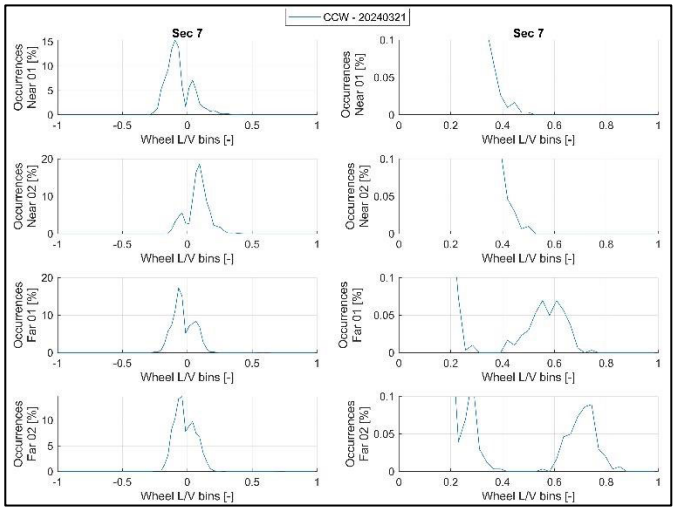


Fig. 6. Example of FAST TPD data at the car level.

the vertical forces on the low rail for lead axles, the horizontal axis shows the train axle sequence, and the vertical axis shows the forces in kips (pounds x 1000).

Another example of TPD data visualization at FAST includes the generation of histograms for each TPD location. Fig. 6 presents the tails of the distribution for the L/V ratio at the wheel for one of the TPD locations. This visualization allowed the identification of single wheel L/V ratio greater than 0.6 on the tangent section, which allowed engineers and operations teams to take a closer look at the corresponding wheel for maintenance intervention.

The design and deployment of Levels 3 and 4 of DT implementation are planned for future developments, beginning with the creation of a DT for a mechanical or track component that uses the existing data collection systems. The selection of the initial component or physical object will be driven by the needs of the FAST operation and the value the component/object adds to ongoing research and testing.

VI. Conclusion

The increasing interest in DTs within the railroad industry presents numerous opportunities for further research and practical applications, especially through initiatives like FAST. While challenges to achieving the full predictive capabilities of the DT remain, the groundwork laid in Level 1 and Level 2 provides a solid foundation for future improvements. Continued efforts to enhance the accuracy and functionality of DTs will be critical in supporting more advanced applications, such as predictive maintenance, and will ultimately contribute to the resilience and efficiency of railroad operations. However, for these technologies to be effectively integrated and used, it will be crucial to establish a unified definition for DTs in the railroad sector or at least within specific areas of application in the railroad industry. Additionally, addressing implementation challenges, including security and interoperability issues, will be essential to realizing the full potential of DTs in enhancing railroad infrastructure and operations.

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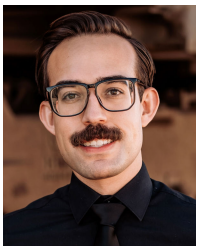
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